

IOT LAB**IV Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE61	ESC	L	T	P	C	CIA	SEE	Total
		-	-	2	1	40	60	100

LIST OF EXPERIMENTS**Introduction to IOT**

1. Introduction to basic electronic components and digital electronic
2. Introduction to sensors and Actuators
3. Introduction to microcontroller
4. Introduction to Arduino IDE

EXPERIMENT PROGRAMS:

1. Blinking of LED with different delays
2. Digital I/O Interface [IR Sensor, PIR Sensor]
3. Analog Interface [ADC, Temperature Sensor]
4. Motor speed And Direction control
5. Serial Communication
6. Wireless Interface –Bluetooth & Wi-Fi Technologies
7. Wireless Control of wheeled robot
8. Smart Home Android App Development

Reference Books:

1. Sylvia Libow Martinez, Gary S Stager, Invent To Learn: Making, Tinkering, and Engineering in the Classroom, Constructing Modern Knowledge Press, 2016
2. Michael Margolis, Arduino Cookbook, Oreilly, 2011

Course Outcomes : Students should be able to:

1. Explain the importance of IOT in aerospace Industry
2. Demonstrate various sensor interfacing using Visual Programming Language
3. Analyze various Physical Components used in IOT lab
4. Demonstrate Wireless Control of Remote Devices
5. Develop an web based application to control the motor